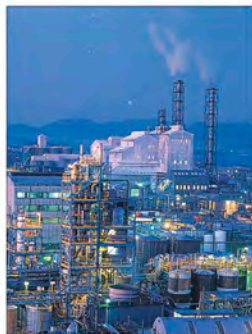


CHEMICAL INDUSTRIES

Soap, fertilizer, and glue are just a few of the useful products of the chemical industries. They are made by combining different substances. Crude oil (unprocessed oil taken straight from the ground) is the main raw material for these industries. The carbon and hydrogen in oil can be made to combine in different ways to make more than half a million things, such as gas, paint, or pills! This manufacturing starts in large factories called refineries.



Oil refinery

Split it up!

Crude oil is split into useful oils inside a distillation tower at a refinery. The oils are separated by being boiled into a gas, and then cooled back into a liquid.

Liquid again

If you put a saucer over a cup of steaming hot liquid, droplets collect on the saucer. The liquid has turned into steam and then cooled back into a liquid when it hit the cold saucer.



Very hot liquid turns into a gas.

Gas cools into a liquid.

Kerosene is the fuel used by airplanes.



Cooling oil drips from the edge of this "saucer" into the tray below.

360°F
(180°C)

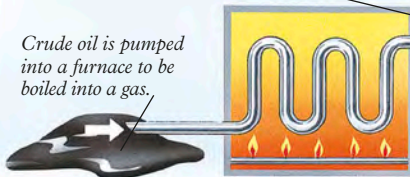
The cloud of crude oil gets cooler and cooler as it wafts up the tower.

Lubricating oil makes machines run smoothly.



At 725°F (385°C), crude oil turns into a gas.

Crude oil is pumped into a furnace to be boiled into a gas.



Distillation tower

230°F
(110°C)